COMPUTER NETWORKS - UE24CS252B

Lab 4: TCP Congestion Window using Wireshark

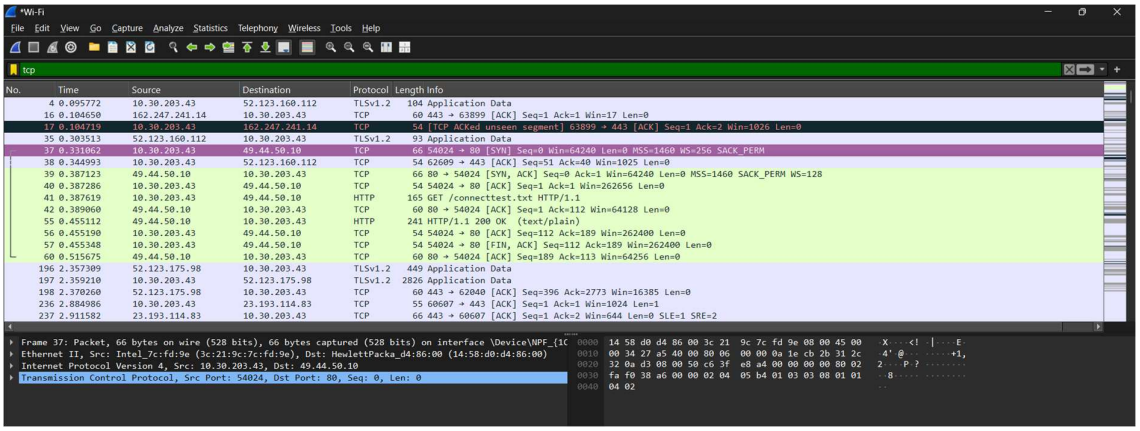
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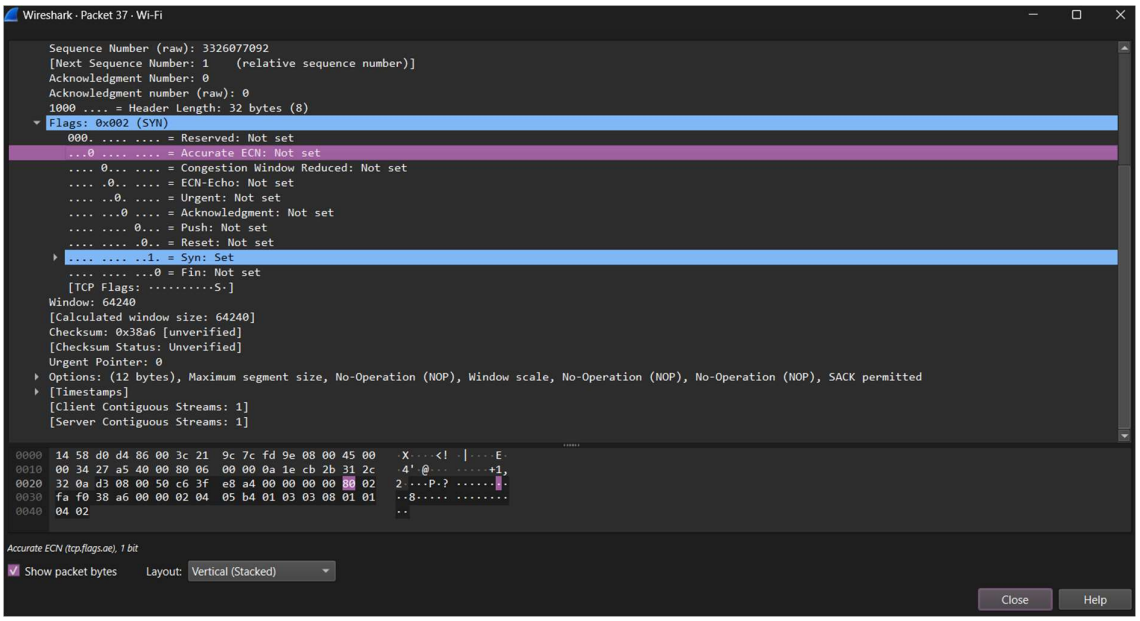
Task A: Capture TCP traffic and plot the Congestion Window

1. Three-Way Handshake Capture



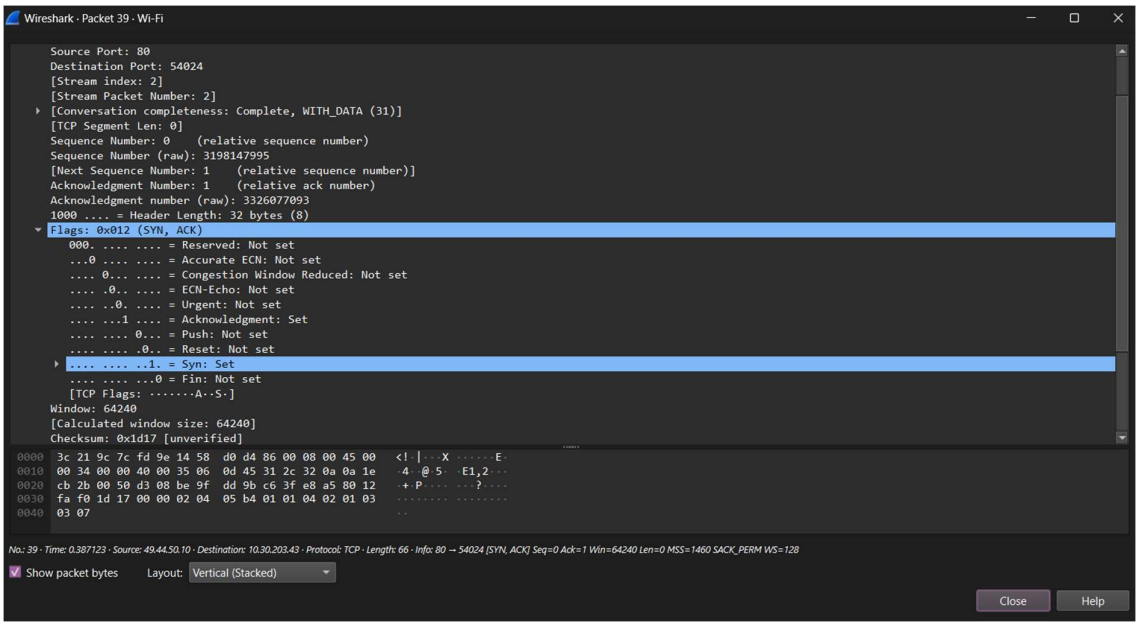
2. TCP SYN Segment Details

- **Question:** What is the sequence number of the TCP SYN segment used to initiate the TCP connection? Additionally, which field or flag indicates that the segment is a SYN packet?
- **Answer:**
  - **Sequence Number:** The relative sequence number is **0**.
  - **Flag:** The **SYN flag** (specifically, the Syn bit set to 1 in the TCP header) tells us that it is a SYN packet.



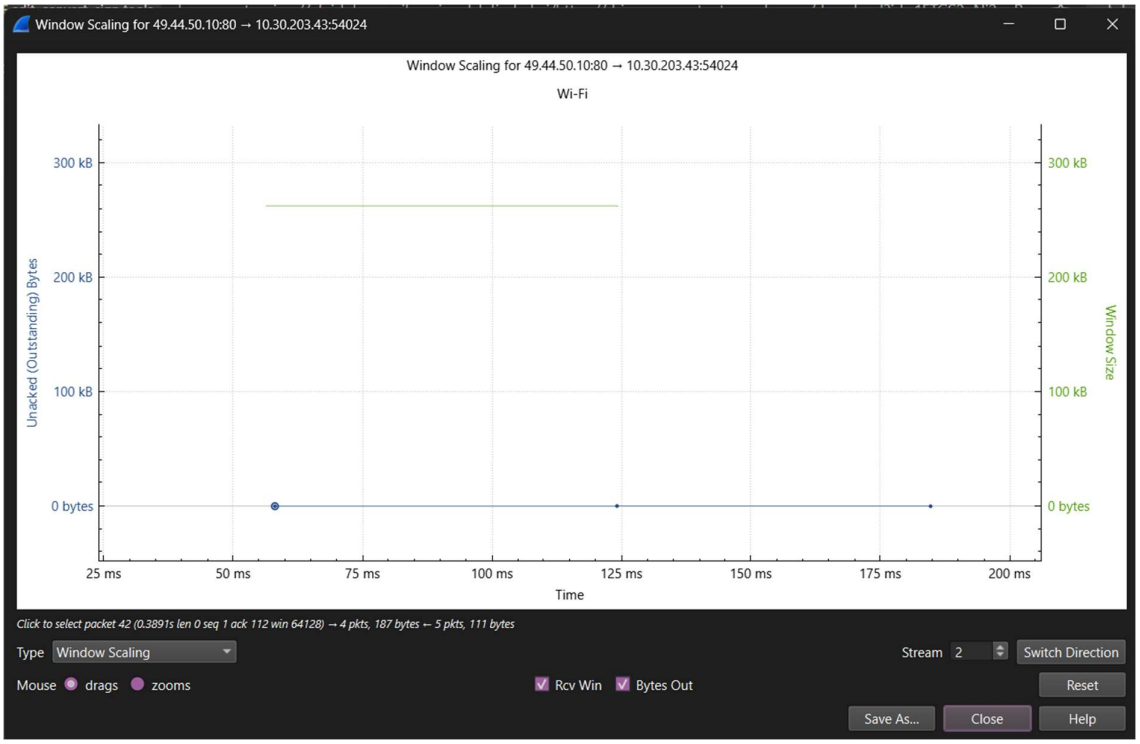
3. SYN-ACK Segment Details

- **Question:** What is the sequence number of the SYN-ACK segment? Which field or flag indicates it is a SYN-ACK packet? What is the value of the Acknowledgement field, and why is it set to that value?
- **Answer:**
  - **Sequence Number:** The relative sequence number is **0**.
  - **Flag:** Both the **SYN flag** and the **Acknowledgment (ACK) flag** are set to 1.
  - **Acknowledgment Value:** The value is **1**.
  - **Reasoning:** It is set to 1 because the server is acknowledging the client's SYN packet (Seq 0) and is now expecting the next byte to be Sequence 1.



4. Congestion Window Plot Analysis

- **Question:** Provide a screenshot of the congestion window. What do the blue and green lines represent?
- **Answer:**
    - **Blue Line/Dots:** Represent the **Unacked (Outstanding) Bytes**, which are the data packets currently "in flight" on the network.
    - **Green Line:** Represents the **Receiver's Window (Rcv Win)**, showing the maximum buffer capacity the receiver has available.



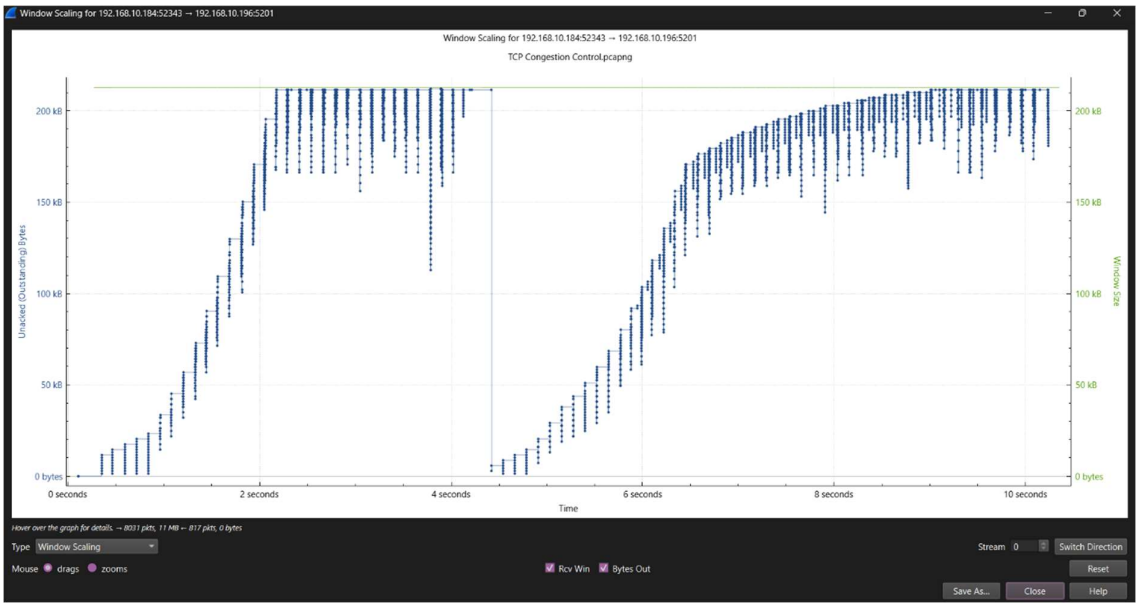
5. Transmission Behavior Inference

- **Question:** What can you infer about TCP transmission behavior (receiver-side or sender-side congestion control) and why?
- **Answer:**
    - **Inference:** The plot indicates **sender-side congestion control**.
    - **Reasoning:** In this plot that is being observed, it can be noticed that the blue dots (data being sent) are significantly lower than the green line (receiver's capacity). This shows that the sender is limiting its transmission rate according to its internal window.

Task B: Analyzing the Congestion Window from the packet capture

1. TCP Phases and Variant Identification

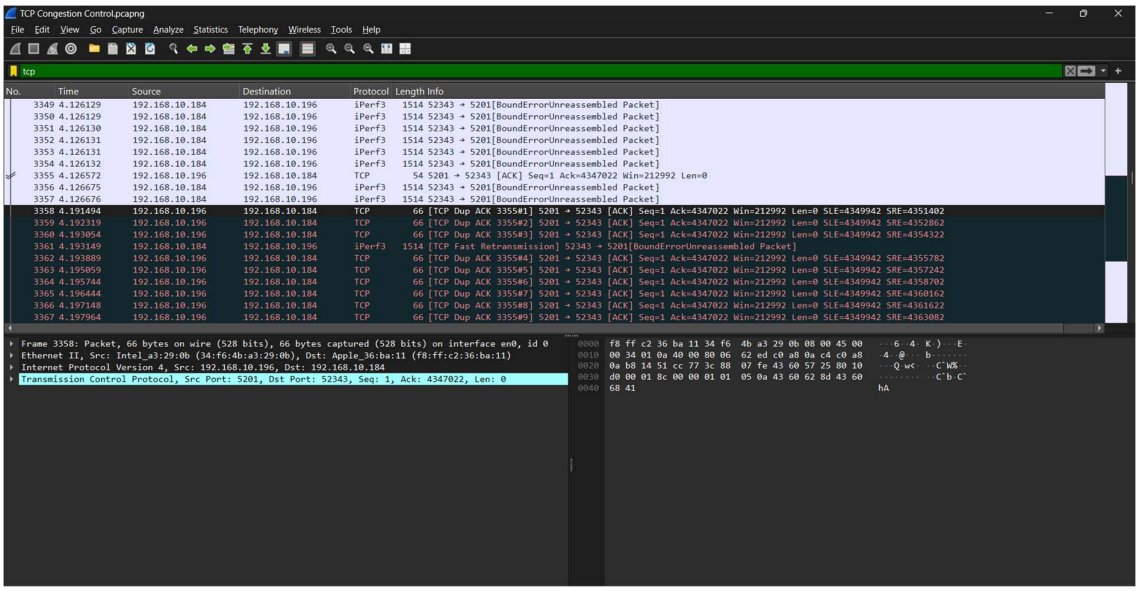
- **Question:** Provide a screenshot of the congestion window and clearly mark the different phases of TCP, such as Slow Start and Congestion Avoidance. Identify which TCP variant (flavor) is being used.
- **Answer:**
    - **Slow Start Phase:** This is seen in the sharp, exponential "step" in window size growth, such as from 0 to 2 seconds.
    - **Congestion Avoidance Phas**e: This is seen in the more gradual, linear diagonal line after a window restarts due to a loss, such as from 6.5 to 10 seconds.
    - **TCP Variant:** TCP Reno is used.
    - **Reasoning:** From the graph, it is seen that the window is divided by two in Multiplicative Decrease when a loss is detected by duplicate ACKs, instead of always dropping to 1.



2. Three Duplicate ACKs and Fast Retransmit

- **Question:** Provide a screenshot that shows three duplicate ACKs. What do you observe in the subsequent packet transmission, and how would you explain this behavior?
- **Answer:**
    - **Observation:** In the packet list, three similar **TCP Dup ACK** packets are received (Frames 3358 to 3360). Immediately after this, a **TCP Fast Retransmission** packet is sent (Frame 3361).

- **Explanation:** This is the "Fast Retransmit" mechanism. The sender receives three duplicate ACKs for the same segment, which implies that the segment has been lost, and it immediately retransmits the segment.

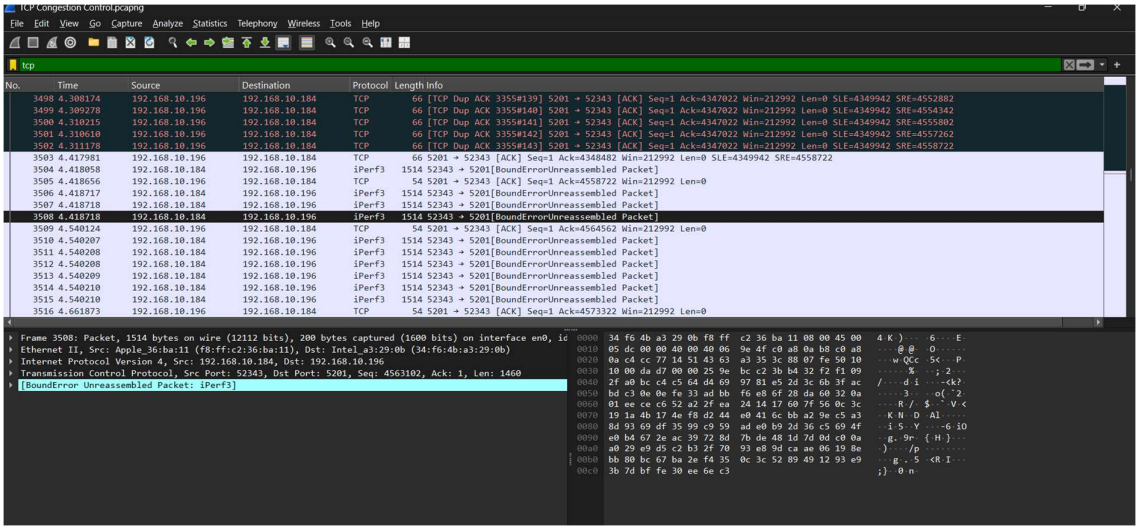


3. Conditions for Window Reduction to 1

- **Question:** Explain under what conditions the TCP congestion window (cwnd) decreases to 1, and describe the network event that causes this behavior.
- **Answer:**
  - **Conditions:** The cwnd decreases to 1 when a **Retransmission Timeout (RTO)** occurs.
  - **Network Event:** The reason for this is that this situation is caused by severe network congestion. In this case, where the network is so congested that even duplicate ACKs are not being sent, the timer is triggered. In order to clear this kind of congestion, the window size is reset to 1 and Slow Start is initiated again.

4. Post-Reduction Packet Observation

- **Question:** Provide a screenshot of the packets observed after the congestion window is reduced to 1. Describe how the congestion window size changes over time.
- **Answer:**
  - **Observation:** After the drop to window size 1 at the 4.4s mark, a **TCP Retransmission** event is observed after a large time gap.
  - **Growth Over Time:** Based on the trend established by the next few packets, the cwnd starts growing **exponentially**. It can be observed that the cwnd doubles with each Round Trip Time as the sender goes back to the **Slow Start** phase to test the available bandwidth again.


TCP Congestion Control.pcapng
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
tcp
No. Time Source Destination Protocol Length Info
3498 4.308174 192.168.10.196 192.168.10.184 TCP 66 [TCP Dup ACK 3355#139] 5201 → 52343 [ACK] Seq=1 Ack=4347022 Win=212992 Len=0 SLE=4349942 SRE=4552882
3499 4.309278 192.168.10.196 192.168.10.184 TCP 66 [TCP Dup ACK 3355#140] 5201 → 52343 [ACK] Seq=1 Ack=4347022 Win=212992 Len=0 SLE=4349942 SRE=4554342
3500 4.310215 192.168.10.196 192.168.10.184 TCP 66 [TCP Dup ACK 3355#141] 5201 → 52343 [ACK] Seq=1 Ack=4347022 Win=212992 Len=0 SLE=4349942 SRE=4555802
3501 4.310610 192.168.10.196 192.168.10.184 TCP 66 [TCP Dup ACK 3355#142] 5201 → 52343 [ACK] Seq=1 Ack=4347022 Win=212992 Len=0 SLE=4349942 SRE=4557262
3502 4.311178 192.168.10.196 192.168.10.184 TCP 66 [TCP Dup ACK 3355#143] 5201 → 52343 [ACK] Seq=1 Ack=4347022 Win=212992 Len=0 SLE=4349942 SRE=4558722
3503 4.417981 192.168.10.196 192.168.10.184 TCP 66 5201 → 52343 [ACK] Seq=1 Ack=4348482 Win=212992 Len=0 SLE=4349942 SRE=4558722
3504 4.418058 192.168.10.184 192.168.10.196 iPerf3 1514 52343 → 5201[BoundErrorUnreassembled Packet]
3505 4.418656 192.168.10.196 192.168.10.184 TCP 54 5201 → 52343 [ACK] Seq=1 Ack=4558722 Win=212992 Len=0
3506 4.418717 192.168.10.184 192.168.10.196 iPerf3 1514 52343 → 5201[BoundErrorUnreassembled Packet]
3507 4.418718 192.168.10.184 192.168.10.196 iPerf3 1514 52343 → 5201[BoundErrorUnreassembled Packet]
3508 4.418718 192.168.10.184 192.168.10.196 iPerf3 1514 52343 → 5201[BoundErrorUnreassembled Packet]
3509 4.540124 192.168.10.196 192.168.10.184 TCP 54 5201 → 52343 [ACK] Seq=1 Ack=4564562 Win=212992 Len=0
3510 4.540207 192.168.10.184 192.168.10.196 iPerf3 1514 52343 → 5201[BoundErrorUnreassembled Packet]
3511 4.540208 192.168.10.184 192.168.10.196 iPerf3 1514 52343 → 5201[BoundErrorUnreassembled Packet]
3512 4.540208 192.168.10.184 192.168.10.196 iPerf3 1514 52343 → 5201[BoundErrorUnreassembled Packet]
3513 4.540209 192.168.10.184 192.168.10.196 iPerf3 1514 52343 → 5201[BoundErrorUnreassembled Packet]
3514 4.540210 192.168.10.184 192.168.10.196 iPerf3 1514 52343 → 5201[BoundErrorUnreassembled Packet]
3515 4.540210 192.168.10.184 192.168.10.196 iPerf3 1514 52343 → 5201[BoundErrorUnreassembled Packet]
3516 4.661873 192.168.10.196 192.168.10.184 TCP 54 5201 → 52343 [ACK] Seq=1 Ack=4573322 Win=212992 Len=0
Frame 3508: Packet, 1514 bytes on wire (12112 bits), 200 bytes captured (1600 bits) on interface en0, id
Ethernet II, Src: Apple_36:ba:11 (f8:ff:c2:36:ba:11), Dst: Intel_a3:29:0b (34:f6:4b:a3:29:0b)
Internet Protocol Version 4, Src: 192.168.10.184, Dst: 192.168.10.196
Transmission Control Protocol, Src Port: 52343, Dst Port: 5201, Seq: 4563102, Ack: 1, Len: 1460
[BoundError Unreassembled Packet: iPerf3]
0000 34 f6 4b a3 29 0b f8 ff c2 36 ba 11 08 00 45 00
0010 05 dc 00 00 40 00 40 06 9e 4f c0 a8 0a b8 c0 a8
0020 0a c4 cc 77 14 51 43 63 a3 35 3c 88 07 fe 50 10
0030 10 00 da d7 00 00 25 9e bc c2 3b b4 32 f2 f1 09
0040 2f a0 bc c4 c5 64 d4 69 97 81 e5 2d 3c 6b 3f ac
0050 bd c3 0e 0e fe 33 ad bb f6 e8 6f 28 da 60 32 0a
0060 01 ee ce c6 52 a2 2f ea 24 14 17 60 7f 56 0c 3c
0070 19 1a 4b 17 4e f8 d2 44 e0 41 6c bb a2 9e c5 a3
0080 8d 93 69 df 35 99 c9 59 ad e0 b9 2d 36 c5 69 4f
0090 e0 b4 67 2e ac 39 72 8d 7b de 48 1d 7d 0d c0 0a
00a0 a0 29 e9 d5 c2 b3 2f 70 93 e8 9d ca ae 06 19 8e
00b0 bb 80 bc 67 ba 2e f4 35 0c 3c 52 89 49 12 93 e9
00c0 3b 7d bf fe 30 ee 6e c3